Advanced Algorithms
Second Programming Project 2026/27
DI – NOVA FCT

The second programming project is completed in teams of **two** students and consists of a theoretical and experimental comparative study of alternative algorithms for **a single** problem. Each group must choose the problem to study and at least two algorithms for the chosen problem. One of the algorithms must be a **randomized** algorithm; the other(s) may not be.

The report should contain the following topics:

- **Problem:** Define the problem and indicate its complexity class or the complexity class of the corresponding decision problem. The proof is not required.
- **Algorithms:** For each (chosen) algorithm, describe the algorithm and analyse its space complexity. Besides, determine:
  - its expected running time, if it is a Las Vegas algorithm;
  - the probability that the answer is incorrect, if it is a Monte Carlo algorithm;
  - its time complexity, if it is a Monte Carlo or a deterministic algorithm.

  Proofs are not required, but a reference for each of them must be provided.
- **Tests:** Characterise the instances used in the experiments, specifying whether they come from available benchmarks or have been created by you.
- **Test Results:** Present and analyse the experimental results, comparing the implemented algorithms. The analysis should take into account execution times.
- **AI Tools:** Specify which websites were consulted and which AI tools were used, for what purposes, and how the information obtained was used to create the source code, the tests, and the report.

Send the report (as a PDF file), the source code, and the input files containing the instances used in the experiments to Margarida Mamede (`mm@fct.unl.pt`). The **deadline** is **November 22nd** (Sunday).